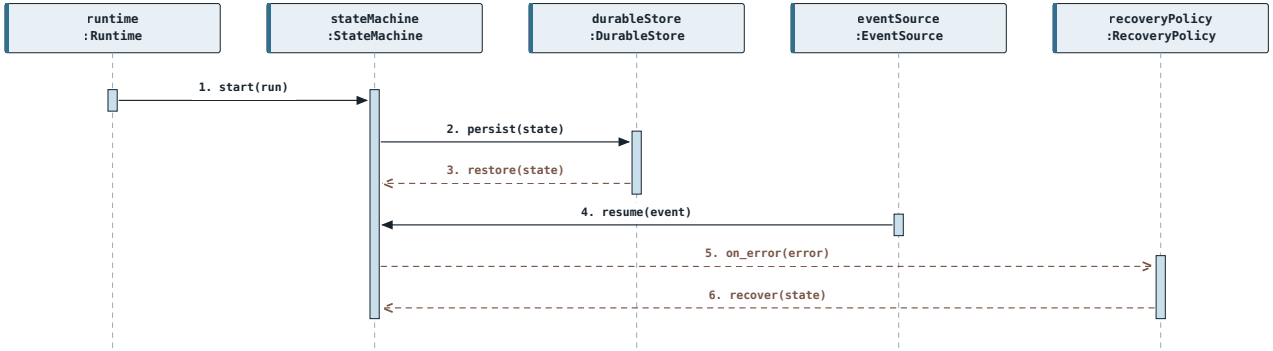
P03 Asynchronous state machine

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Make waiting and resumption explicit.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	A harness run may pause for external work or user input.
TRIGGER	An event requests suspension or arrives to resume a paused run.
STRUCTURE	Enumerated states, legal transitions, event identity, and resumption state.
EFFECT	A resumed run follows only the legal transition and preserves durable state.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	State can be serialized or retained and incoming events can be identified.
ENSURES	Accepted events move the machine to a defined state and duplicate events are handled.
INVARIANT	No transition crosses from a disallowed state or executes twice after replay.
VIOLATION	Invalid, duplicate, or interrupted events leave a recoverable state or raise explicitly.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.